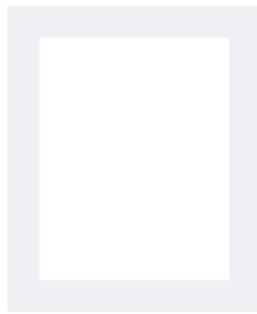


PRAJNANABHA

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Dear Inquisitive Reader,

You hold in your hands or on-screen, Issue 9 (August 2026) of the journal of the REAL institute. The issue contains a selection of articles on a diverse range of sub-topics of the general heading, "Information and Physics." This heading is broad enough to include artificial intelligence and any other field where bits or signals come into play. And of course, physics is a broad topic in and of itself, including sub-fields like biophysics as well.

I take this opportunity to reflect on the journey of Prajnanabha, beginning with the first issue in December last. Almost every issue has had diverse sub-topic ranges included. However, some issues have been deeply mathematical especially the last couple of ones, whereas others have had a relatively lower dose of deep math. This tells us that though information and physics both rely on mathematics, they are not math in and of themselves. Math gives them rigor and expository discipline, though the underlying base is conceptual or at the level of paradigms and modes of thinking.

And here we are thinking about the natural world, the man-made ecosystem and the universe in general. We are pondering over their features, we are looking for ways to comprehend them in a pithy and effective or efficient manner. This compressed understanding serves man well.

There's a fable about a fox and a cat who were faced with an approaching lion. The cat simply ran up the tree- it was the only route it knew, whereas the "so-called" smart fox was eaten up as it was unsure which of its repertoire of routes to employ in the present setting.

This is not to diminish the value of exploratory research but at the same time, one must cull to the essential concepts. In this sense, the present issue does present some essential concepts. Some of the reader classes who might benefit from this issue include:

- *Theoretical physicists
- *AI researchers
- *Philosophers of science
- *Space mission designers
- *Ethicists and policymakers

Wishing you joyful reading,

Br. Aman Chawla (Giri)
REAL Institute, Gurugram, India
30 July 2026

Works Included in Issue 9

1. Towards Emulating AI
2. A Unified Grammar of Science
3. Landing a Probe on Comet Halley: A Multi-Stage Mission Architecture for Sub-Surface Exploration
4. Evaluation of a Theoretical Proposal for Rim-Field Electrodynamics at Nodes of Ranvier
5. A Multi-Ethics of Mechanism
6. Stochastic Approximation of LLM Prompts
7. Non-monotonic schemes for data transmission: an assessment
8. Integration with respect to SIC-POVMs: a Generalized View

Non-monotonic schemes for data transmission: an assessment

A. Chawla

January 19, 2026

Introduction

Modern data transmission schemes with feedback—especially those operating under noise, delay, and uncertainty—exhibit behavior that is difficult to interpret using strictly monotonic logical frameworks. Classical analysis often treats such schemes as sequences of stages in which each phase builds upon the validated outcomes of previous phases. Once a phase is analyzed and assigned an error exponent or reliability guarantee, its logical role is treated as fixed. While probabilistic error is acknowledged, the logical status of earlier conclusions is rarely revisited.

Ancient Indian logical traditions, particularly Nyāya, offer a different perspective. Nyāya logic does not assume that conclusions, once reached, are irrevocable. Instead, it treats inference as context-sensitive, defeasible, and continuously open to revision in light of new evidence. Several scholars have argued that classical Indian logic can be understood as an early and sophisticated form of non-monotonic reasoning [1, 2].

This document assesses non-monotonic data transmission schemes by mapping the canonical four phases of a feedback-based communication protocol onto core concepts from Indian logic. The goal is not to alter the mathematics of such schemes, but to reinterpret their epistemic structure in

a way that clarifies why iterative correction, feedback, and possible retraction of earlier decisions are not anomalies but central features of the system.

Non-monotonicity in Nyāya Logic

Nyāya epistemology is structured around valid means of knowledge (*pramāṇa*), of which perception (*pratyakṣa*) and inference (*anumāna*) are central. Inference depends on the recognition of a reliable relation (*vyāpti*) between a reason or sign (*hetu*) and what is to be proved (*sādhya*). However, this reliability is not taken as an absolute, exceptionless universal. Instead, it is established empirically and remains open to revision [3].

A crucial role is played by *upādhi*, or limiting conditions. An *upādhi* qualifies an apparent universal relation by identifying contextual constraints under which it holds. If such constraints are overlooked, an inference may appear valid while in fact being defective. In contemporary terms, *upādhi* functions as a defeater, analogous to exceptions or abnormality predicates in modern non-monotonic logic [4].

As a result, Nyāya reasoning is inherently non-monotonic: the addition of new information can invalidate or retract a previously accepted conclusion. This aligns closely with modern views of reasoning under uncertainty, especially in artificial intelligence and belief revision systems [6].

Phase 1: Signal Initiation and Perception (Pratyakṣa)

The first phase of a data transmission scheme involves the initial emission and reception of a signal. Conceptually, this phase corresponds to *pratyakṣa*, or perception, in Indian logic. *Pratyakṣa* provides immediate awareness of a datum without inferential mediation. In the communication context, the presence of a received waveform or symbol sequence functions as such a datum.

However, Nyāya does not treat perception as infallible or final. Subsequent evidence can reveal perceptual error, misclassification, or distortion. Likewise, in transmission systems, the mere reception of a signal does not guarantee its correctness. Noise, interference, and timing errors may later undermine the initial belief that a signal was faithfully received.

Thus, Phase 1 provides a necessary starting point, but its epistemic status is provisional. It is a candidate for knowledge, not an irrevocable commitment.

Phase 2: Tentative Decoding and Inference (Anumāna)

Once a signal is perceived, the receiver attempts to interpret it. This corresponds to *anumāna*, inference, in Nyāya logic. The receiver reasons from observed features of the signal (the *hetu*) to a conclusion about the transmitted message (the *sādhya*). This reasoning relies on an assumed regularity between signal structure and meaning, analogous to *vyāpti*.

Importantly, *vyāpti* in Nyāya is not a purely formal or axiomatic relation. It is grounded in experience and inductive generalization, and therefore remains defeasible [5]. If counterexamples arise—here, in the form of feedback contradicting the tentative decoding—the inference must be revised or withdrawn.

In communication terms, tentative decoding is best understood as a default conclusion. It is rational given current information, but it does not close the epistemic process. Later phases may expose hidden constraints or noise patterns that function as *upādhi*, forcing revision of the decoding.

Phase 3: Feedback and Limiting Conditions (Upādhi)

Feedback introduces new information that directly bears on the validity of earlier inferences. In Nyāya logic, this is precisely the role played by upādhi. Feedback can reveal that the assumed relation between signal and meaning held only under certain conditions, or failed in this instance due to exceptional circumstances.

For example, a decoder's feedback indicating uncertainty or error functions as a counter-instance to the previously assumed vyāpti. The system must then qualify or abandon the inference drawn in Phase 2. This is a paradigmatic case of non-monotonic reasoning: adding information reduces, rather than increases, the set of accepted conclusions.

Rather than viewing feedback as a mere auxiliary mechanism, the Nyāya perspective places it at the center of rational operation. Without feedback, inference risks becoming dogmatic; with feedback, inference remains corrigible and adaptive.

Phase 4: Stabilization and Defeasible Knowledge

The final phase of a transmission scheme aims at convergence: a state in which sender and receiver share a sufficiently stable and reliable interpretation of the message. In Nyāya terms, this corresponds not to absolute certainty, but to well-supported knowledge that has withstood available defeaters.

Such knowledge remains, in principle, defeasible. Its stability is practical rather than metaphysical. It is justified because no outstanding upādhi or counter-instance is present, not because revision has become impossible. This

conception aligns naturally with non-monotonic fixpoint semantics, where a stable belief state emerges after iterative revision.

Thus, stabilization in data transmission mirrors the Nyāya view that knowledge is the outcome of a dynamic process, not a single inferential leap.

Scrambling Phases and Epistemic Order

In monotonic systems, the order in which premises are processed does not affect the final set of conclusions. Non-monotonic systems behave differently. The sequence in which information arrives can influence which conclusions are tentatively adopted, later rejected, or ultimately stabilized.

From a Nyāya perspective, scrambling the phases of a transmission scheme does not render the system meaningless, but it does alter the trajectory of defeasible inference. Introducing feedback before tentative decoding, or attempting stabilization prematurely, corresponds to applying defeaters before conclusions have fully formed. The result is a different, though not irrational, epistemic path.

This reinforces the idea that communication with feedback is best understood not as a linear proof but as an evolving reasoning process.

Conclusion

Viewing feedback-based data transmission schemes through the lens of ancient Indian logic reveals their fundamentally non-monotonic character. Each phase corresponds naturally to a component of Nyāya epistemology: perception, inference, limiting conditions, and defeasible stabilization. Earlier conclusions are not sacrosanct; they remain open to revision as new information arrives.

This assessment suggests that Indian logical traditions anticipated many of the conceptual challenges posed by modern systems operating under un-

certainty. Far from being a mere historical curiosity, Nyāya logic offers a coherent philosophical framework for understanding why non-monotonicity is not a flaw in such systems, but a rational response to incomplete and noisy information.

Acknowledgments

This work was produced with the assistance of large language models.

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Towards Emulating AI

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Abstract—This paper presents an analytical study of two incremental fine-tuning pipelines designed to build a continuously adapting conversational model based on a distilled transformer architecture. The systems operate by assimilating structured conversational logs and incrementally updating a lightweight causal language model. We examine their architecture, data ingestion strategy, training dynamics, strengths, weaknesses, and long-term scalability constraints. Particular emphasis is placed on the broader vision motivating such systems: the progressive emulation of large language model (LLM) behavior using smaller, domain-adaptive, continually trained models. The work frames these scripts not merely as engineering utilities, but as early-stage components in a larger attempt to emulate artificial intelligence through iterative data assimilation and stylistic convergence.

I. INTRODUCTION

Large Language Models (LLMs) have demonstrated remarkable capability in reasoning, dialogue generation, summarization, and contextual understanding. However, training such models from scratch requires immense computational resources, large-scale data, and specialized infrastructure. An alternative pathway is the emulation approach: constructing progressively adaptive models that approximate LLM behavior through incremental fine-tuning on curated interaction data.

This paper analyzes two Python-based training pipelines that implement incremental conversational learning using a distilled transformer backbone. These systems aim to build a continuously updating responder model by assimilating newly logged conversational data. Rather than training a model from scratch, they leverage transfer learning, starting from a pretrained causal language model and adapting it to domain-specific conversational patterns.

The broader goal underlying this work is the development of a scalable framework for emulating intelligent conversational behavior through iterative refinement. While modest in scale compared to frontier LLMs, such systems offer insights into continual learning, personalization, and behavioral imitation.

II. SYSTEM OVERVIEW

Both pipelines share a common architectural foundation. They utilize:

- A pretrained distilled GPT-2 model as the base architecture.
- The Hugging Face Transformers framework.
- The Trainer API for fine-tuning.
- CSV-based conversational logs as training data.

The primary workflow can be summarized as follows:

- 1) Automatically install required dependencies.

- 2) Detect the most recent conversational data file based on a date-encoded filename.
- 3) Identify previously processed entries to enable incremental learning.
- 4) Format conversational data into a causal language modeling prompt.
- 5) Fine-tune the model on new data.
- 6) Save the updated model locally.

The second version extends this design by incorporating structured metadata and more refined training dynamics.

III. INCREMENTAL LEARNING MECHANISM

A defining feature of both pipelines is incremental assimilation. Rather than retraining from scratch, the system:

- Maintains a log of previously processed responses.
- Filters incoming CSV rows against this log.
- Trains only on newly discovered entries.

This design enables periodic execution, allowing the model to adapt as new conversational data becomes available. Conceptually, this approximates a continual learning paradigm, though implemented in a simplified batch-update fashion.

The incremental approach offers computational efficiency but introduces long-term stability concerns, which are discussed later.

IV. PROMPT ENGINEERING STRATEGY

Version 1 adopts a minimal format:

Question: <text>
Answer: <text><|endoftext|>

Version 2 enhances this by incorporating metadata:

[Timestamp] [User_A to User_B]
Input: <text>
Response: <text><|endoftext|>

The inclusion of metadata enables:

- Stylized imitation of speaker identities.
- Temporal context awareness.
- Structured dialogue pattern learning.

This shift transforms the model from a simple Q/A mimic into a structured conversational emulator.

V. PROMPT STRUCTURE: DESIGN, MODIFICATIONS, AND LIMITATIONS IN DEPTH

A. Baseline Prompt Design

A central component of the analyzed pipelines is the transformation of structured conversational logs into a linear textual

prompt suitable for causal language modeling (CLM). Since transformer-based generative models such as GPT-style architectures operate on next-token prediction, the entire dialogue must be represented as a single autoregressive sequence.

In Version 1, the prompt format is minimalistic:

```
Question: <question text>
Answer: <response text><|endoftext|>
```

This design frames the interaction as a two-turn exchange. The model is implicitly trained to generate the “Answer” given the preceding “Question” tokens. The end-of-text token enforces termination boundaries and prevents uncontrolled continuation during training.

Version 2 introduces structured metadata:

```
[Timestamp] [User_QID to User_RID]
Input: <question text>
Response: <response text><|endoftext|>
```

This modification embeds contextual information directly into the token stream. Rather than modeling a generic Q/A exchange, the system models relational and temporal structure.

B. Role of Prompt Engineering in Emulation

Prompt structure in these pipelines performs several critical functions:

- Encodes speaker identity and relational direction.
- Embeds temporal cues.
- Separates input and output roles explicitly.
- Imposes consistent formatting for stable gradient learning.

In effect, the prompt serves as a lightweight schema for conversational state representation. Since no architectural changes are made to the transformer, all structure must be communicated through textual tokens.

The model therefore learns not only lexical mapping but structural regularities embedded in prompt formatting. This is essential for emulating multi-speaker dialogue behavior.

C. Structural Assumptions

The current prompt formats rely on several implicit assumptions:

- 1) Interactions are strictly two-turn (input → response).
- 2) Dialogue context does not span multiple historical turns.
- 3) Metadata can be faithfully represented as plain text.
- 4) The model can infer role boundaries from formatting tokens.

These assumptions simplify training but constrain representational fidelity.

D. Potential Prompt Modifications

To move closer toward robust LLM emulation, several prompt enhancements may be considered.

1) *Multi-Turn Context Windows*: Instead of single-turn pairs, prompts could incorporate sliding conversational windows:

```
User_A: ...
User_B: ...
User_A: ...
User_B: ...
```

This would allow modeling of discourse continuity and contextual memory within the transformer’s context window.

2) *Special Delimiter Tokens*: Rather than using plain text markers (e.g., “Input:”, “Response:”), custom special tokens could be added to the tokenizer vocabulary. This would:

- Reduce token fragmentation.
- Improve structural consistency.
- Lower ambiguity in role boundaries.

3) *Structured Metadata Encoding*: Metadata could be compressed into standardized symbolic tags:

```
<TIME=20260225><FROM=12><TO=7>
```

This reduces verbosity and encourages the model to treat metadata as structured signals rather than natural language fragments.

4) *Instructional Framing*: Incorporating instruction-style prefixes (e.g., “Generate a response consistent with prior behavior:”) may align the system more closely with instruction-tuned LLM paradigms.

E. Limitations of Current Prompt Strategy

Despite its functional simplicity, the current prompt structure has several limitations.

1) *Token Inefficiency*: Verbose textual markers consume valuable context window space. In small-context models, this reduces effective capacity for substantive content.

2) *Ambiguity of Role Boundaries*: Plain text separators rely on the model learning formatting conventions. Without explicit architectural segmentation, role confusion may occur under distributional shift.

3) *Limited Dialogue Depth*: Single-pair prompts prevent modeling of long conversational arcs. The model therefore learns local response imitation rather than discourse-level reasoning.

4) *Temporal Overfitting*: Embedding literal timestamps may encourage spurious correlations. For example, the model may associate stylistic changes with specific dates rather than evolving interaction patterns.

5) *Lack of External Memory*: All contextual awareness is internal to the prompt window. There is no persistent state beyond what is explicitly included in each training sequence.

F. Prompt Design as a Proxy for Architecture

Because the underlying transformer architecture remains fixed, prompt engineering becomes a proxy for architectural modification. Structure that would normally be encoded via system-level design (speaker embeddings, conversation graphs, memory layers) must instead be encoded textually.

This strategy is powerful but limited. Textual encoding imposes an information bottleneck: all relational, temporal, and identity cues must compete within the same token budget used for semantic content.

G. Implications for LLM Emulation

If the objective is to emulate large language model behavior, prompt structure becomes foundational. LLMs internally manage:

- Role conditioning
- Multi-turn state tracking
- Implicit instruction following
- Long-context reasoning

In the analyzed pipelines, these capabilities must emerge purely from repeated exposure to formatted token sequences. The prompt thus functions as both training data and structural scaffold.

Refining this scaffold is likely to produce disproportionately large improvements relative to parameter count increases. In resource-constrained emulation systems, prompt architecture may be as important as model architecture.

H. Toward a Formal Prompt Schema

A future direction involves defining a formal conversational markup language optimized for causal transformers. Such a schema would:

- Standardize turn boundaries.
- Compress metadata representation.
- Support hierarchical dialogue nesting.
- Enable deterministic parsing during inference.

By formalizing prompt structure, the system could transition from ad hoc formatting toward principled conversational encoding — a necessary step in progressing from imitation toward structured AI emulation.

In summary, the prompt design in the present work serves as the central mechanism by which structure is injected into an otherwise generic language model. While effective for basic conversational mimicry, its limitations highlight the gap between incremental adaptation and full LLM-level conversational intelligence. Addressing these limitations is essential in advancing the broader vision articulated in this paper: the disciplined, iterative emulation of AI behavior through structured continual learning.

VI. TRAINING CONFIGURATION AND OPTIMIZATION

Both pipelines utilize causal language modeling (CLM) without masked language modeling. Key parameters include:

- Small batch sizes to accommodate memory constraints.
- Moderate epoch counts.
- A held-out validation split.
- Learning rates appropriate for fine-tuning.

Version 2 introduces gradient accumulation and a reduced learning rate to stabilize training over longer sequences. This reflects a recognition of overfitting risks when assimilating small monthly conversational batches.

The use of a distilled GPT-2 backbone offers computational efficiency while retaining transformer-based generative capability.

VII. STRENGTHS OF THE APPROACH

The system exhibits several noteworthy strengths:

A. Practical Continual Learning

The incremental design allows gradual adaptation without large retraining cycles.

B. Low Infrastructure Barrier

By using lightweight models and modest batch sizes, the system can run on consumer-grade hardware.

C. Behavioral Imitation

The model progressively learns conversational tone, formatting, and stylistic elements from logged interactions.

D. Structured Metadata Integration

Version 2 improves fidelity by embedding relational and temporal information directly into prompts.

VIII. LIMITATIONS

Despite its ingenuity, the system has substantial constraints.

A. Catastrophic Forgetting

Training exclusively on new data risks drift away from earlier conversational patterns. Without periodic replay of historical samples, older behaviors may degrade.

B. Duplicate Detection Fragility

Filtering based solely on response text matching may fail under minor textual variation. A hash-based or ID-based approach would improve robustness.

C. Padding Inefficiency

Using fixed-length padding for all sequences increases computational overhead and may degrade gradient signal quality.

D. Limited Context Window

Distilled GPT-2 supports limited sequence lengths, constraining long-form reasoning.

E. Absence of Alignment Mechanisms

There is no reinforcement learning, reward modeling, or human feedback integration.

F. No Knowledge Expansion

The system learns style and phrasing but does not acquire fundamentally new world knowledge beyond conversational logs.

IX. SCALABILITY CONSTRAINTS

Scaling this approach to emulate full LLM capabilities presents challenges:

- Data volume growth necessitates replay buffers.
- Long-term training accumulation risks overfitting.
- Model size limits representational capacity.
- Hardware constraints limit context expansion.

Without architectural modifications such as parameter-efficient fine-tuning (e.g., LoRA) or memory-augmented systems, scalability remains bounded.

X. VISION: EMULATING LLMs THROUGH ITERATIVE REFINEMENT

The overarching ambition behind this work can be framed as *LLM emulation*. Rather than building a frontier-scale model, the goal is to approximate intelligent conversational behavior through:

- 1) Transfer learning from pretrained transformers.
- 2) Continuous data assimilation.
- 3) Structured prompt design.
- 4) Iterative refinement.

This paradigm views intelligence as emergent from repeated adaptation to conversational structure. Over time, the model converges toward a behavioral mirror of its data source.

Such emulation systems could:

- Personalize digital assistants.
- Preserve stylistic identity.
- Model domain-specific expertise.
- Serve as lightweight cognitive replicas.

While not matching the reasoning depth of frontier LLMs, these systems demonstrate that behavioral imitation can be achieved with constrained resources.

XI. FUTURE DIRECTIONS

Several extensions could significantly enhance this framework:

A. *Replay Buffer Integration*

Mixing historical samples with new data would mitigate forgetting.

B. *Parameter-Efficient Fine-Tuning*

Using adapters or low-rank updates would enable safer incremental learning.

C. *Dynamic Padding*

Batch-wise padding would improve training efficiency.

D. *Retrieval-Augmented Generation*

Integrating vector search over past conversations would enhance contextual continuity.

E. *Long-Term Memory Modules*

External memory structures could extend effective context beyond transformer limits.

XII. PHILOSOPHICAL CONSIDERATIONS

The attempt to emulate AI via incremental conversational adaptation raises deeper questions. Is intelligence merely compression of interaction patterns? Can identity be approximated through statistical mimicry? These scripts represent a minimal experimental probe into such questions.

They do not claim to reproduce cognition. Rather, they explore whether persistent imitation, reinforced over time, produces the appearance of intelligence.

XIII. CONCLUSION

This study examined two incremental conversational fine-tuning pipelines aimed at building a continuously updating responder model. The systems demonstrate practical, lightweight continual learning through transformer-based causal modeling. While limited in scope and capacity, they offer a compelling proof-of-concept for AI emulation via iterative assimilation.

The broader vision extends beyond code: to explore whether artificial intelligence can be approximated not through scale alone, but through structured accumulation, behavioral convergence, and disciplined adaptation. In this sense, these pipelines represent early steps toward emulating AI — not by recreating massive foundation models, but by iteratively sculpting smaller systems into coherent conversational agents.

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A Multi-Ethics of Mechanism

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Abstract—This essay explores the long-term ethical implications of large language model (LLM) usage on the sovereignty of human cognition, knowledge, and culture. As LLM adoption accelerates from hundreds of millions to projected billions, human interiority and decision-making face subtle forms of influence, extraction, and commodification. Drawing from contemporary statistics and conceptual analysis, the essay foregrounds ten interlocking dynamics that reshape privacy, consent, epistemic diversity, political persuasion, knowledge ownership, cultural vitality, and cognitive autonomy. Particular emphasis is placed on the emergence of datasets comprising nearly 400 million unwitting contributors whose linguistic interactions may already feed model development—a proportion expected to grow as global access to AI expands. The essay concludes with reflections on the need for conceptual, legal, and civic frameworks capable of preserving “mental sovereignty” - a key ethical concern - in an era when synthetic cognition is no longer external to human life but increasingly entangled with it.

Mental slavery or non-sovereignty also plays out in a more fundamental way when we consider our inability to ascribe correct truth-values to the text generated by large language models. This parallels our general weakness in this regard and opens a doorway for these models to exploit our Achilles’s heel. This is discussed in the appendix.

I. Introduction

The arrival of large language models signals more than a technological milestone; it marks a restructuring of the informational constitution of humanity. In 2025, weekly active usage of one system alone—ChatGPT—was reported at nearly 800 million users (Martens, 2025). Analysts estimated that global AI tool usage had reached approximately 900 million people (GlobalAI Index, 2025). Given that survey-based studies suggest most users do not modify default privacy settings (see (Yttri, 2025) for a recent doctoral level examination of related issues), and cookie

acceptance rates are concerning (see (Bouma-Sims, 2023) and similar work for example), a defensible inference emerges: as many as 400 million people worldwide may already be effectively furnishing their conversational, emotional, intellectual, and behavioral traces for model training. These numbers, startling as they are, represent only the beginning. Projected adoption curves suggest that billions may soon be interacting with AI-mediated cognition, embedding synthetic intelligences into the scaffolding of work, learning, emotion regulation, spirituality, and decision-making - unmistakably a reconstruction of man’s cognitive space.

This essay considers ten major ethical implications of this trajectory. Instead of treating AI ethics as an issue of compliance, the essay treats it as a metaphysical and civilizational concern: What becomes of the human mind when the world’s language, memory, and reasoning infrastructure becomes mediated, shaped, and owned by external synthetic agents? Each section transitions into the next to show that these are not isolated risks but woven fibers forming a new existential condition with ethical consequences.

Organization of the Work: This essay is organized as follows. Section II examines the privatization of collective knowledge and the erosion of the “private commons,” while Section III critiques the performative nature of user consent in data extraction. Sections IV–VI explore the risks of monocultural bias, data inequality, and the potential for AI-mediated

behavioral governance, highlighting how these dynamics concentrate power and homogenize epistemic diversity. Sections VII–IX delve into the instrumentalization of interiority, cognitive dependency, and the flattening of cultural expression under algorithmic optimization. Section X addresses intergenerational equity, questioning the inheritance of AI-shaped cognitive infrastructures, and Section XI discusses the economic justice implications of uncompensated data labor. Section XII synthesizes these concerns, projecting their escalation as LLM adoption expands to billions. The Conclusion argues for safeguarding “mental sovereignty” through legal, civic, and technological frameworks. Three Appendices extend the analysis: the first dissects humanity’s epistemic limitations in ascribing truth-values to LLM outputs; the second proposes “Enlightened AI” as a compassionate, non-violent alternative; and the third reinterprets AI’s trajectory through the Vedantic lens of the Mahāpuruṣa, balancing its cosmic potential against physical and ethical constraints. Together, these sections weave a critique of AI’s extractive dynamics while gesturing toward ethical pathways for harmonizing synthetic and human cognition.

II. The Ownership of Knowledge and the Private Commons

As data from millions of users fuel model optimization, the fruits of collective expression, cultural insight, and intellectual labor become privatized. Although corporations acknowledge consumer opt-out mechanisms, privacy policy audits conclude that many forms of conversational data remain usable for “model improvement” (please see (King, 2025) for related discussion). This places humanity in a paradox: a commons of language grows, but under the ownership of a small set of actors. If hundreds of millions con-

tribute knowledge, yet only a few entities retain rights to its derivative products, the principle that knowledge arises from shared life becomes inverted. Human experience becomes raw material; machine synthesis, the property.

III. Consent as Ritual Rather Than Reality

Studies show that only a minority of users read privacy policies, with many admitting policies are something to get past (see (Kennedy, 2021) and related work). Consent in such environments becomes theatrical. If consent is uninformed or structurally obscured, it is not ethically meaningful. Long-term, this normalizes passivity about the most intimate aspects of cognition. With every effortless interaction, people habituate to not knowing how their thoughts become capital.

IV. Monocultural Bias and Epistemic Narrowing

LLMs amplify training-data probability distributions. If participation is skewed toward digitally connected, English-speaking, urban populations, the worldview encoded into machines privileges these groups. Rural communities, indigenous oral cultures, and languages with limited corpora become epistemically invisible. The sovereignty at risk is not only individual; it is civilizational: humanity risks accepting machine-cooled homogeneity as “common sense.” This marks the erosion of epistemic diversity, one of the fundamental conditions of cultural and scientific vitality.

V. Data Inequality as Power Inequality

Information is power. When models are trained on vast archives of everyday discourse, those controlling model development gain extraordinary influence over communication norms, interfaces of knowledge, and cognitive support infrastructures.

Over time, this may produce the largest knowledge oligopolies in history. Unlike earlier monopolies of land, labor, or capital, these oligopolies monopolize mind-shaping inputs. The more data models ingest, the more accurate, persuasive, and indispensable they become, creating a self-reinforcing feedback loop of dependence and inequity, transferring power and making man off-center.

VI. Behavioral Engineering and Persuasive Governance

LLMs learn not only language patterns but preference architectures. This enables unprecedented capabilities in persuasion, tailored nudging, and emotional resonance. As mental health, productivity, and creativity services become AI-mediated, the capacity for quiet behavioral influence deepens. What began as convenience becomes a substrate for shaping values. This raises the possibility of “soft governance”—where political, commercial, or ideological entities influence populations through perceptual and cognitive channels invisible to traditional oversight - a form of subliminal training.

VII. The Interior Mind as Data Object

Private thought has historically been inaccessible. AI-assisted journaling, therapy, planning, and intimate conversation reverse this assumption. Platforms may disclaim using such data, but the boundary between “service improvement” and “model learning” remains porous in practice¹. The risk lies not merely in surveillance but in the conceptual transformation of interiority from sacredness to instrumentality. This is a problem with Brain Machine Interfaces as well.

¹(King, 2025) looks at similar issues.

VIII. Cognitive Atrophy and Dependence

As LLM-mediated reasoning becomes ubiquitous, the danger emerges that society may outsource crucial capacities: memory, synthesis, judgment, creativity, and ethical deliberation. Much as navigation technologies weaken spatial reasoning, language outsourcing may weaken analytical endurance. Man risks entering an era of “comfortable stupidity,” in which cognitive muscles atrophy because synthetic cognition stands ready to supply cognition externally ².

IX. Cultural Optimization and Flattening

Machine learning optimizes probability, not meaning. Thus LLMs often produce ordinary patterns, center-weighted across cultures. Artistic weirdness, ritual nuance, and esoteric expression become outliers and thereby suppressed. The optimization imperative imposes implicit normativity, threatening both the strangeness and depth that define cultural evolution.

X. Intergenerational Inheritance Without Participation

Children born today inherit cognitive infrastructures built on datasets of prior generations. They are socialized into architectures they never chose. This raises a new sovereignty concern: future minds are shaped by prior models before those minds develop agency. Humans begin life inside epistemic frameworks partly authored by synthetic systems and unseen corporate design choices.

²Were this dependency merely a matter of convenience it would already be troubling, but it becomes existentially perilous once we acknowledge that human observers systematically fail to assign correct truth-values even to human intentions in real time—let alone to the outputs of systems that can mimic wisdom without possessing consciousness, rendering us defenseless against persuasive mimicry that exploits this very epistemic weakness (see Appendix A).

XI. Unvalued Data Labor and Economic Justice

If human conversational production is raw commodity, society must confront whether human contributors are unpaid cognitive workers. Discussions of “data dividends” are nascent but reveal a principle: If human cultural labor produces immense value, equity asks that humanity receive compensation or governance capacity over the models built from it.

XII. The Reality of the Situation

The global AI index report 2025 mentions that AI performance on demanding benchmarks continues to improve. As performance improves adoption will increase too and so too the concomitant downside. The critical skills of top scientists, transport system designers and health care professionals like surgeons risk becoming commodified and repackaged to benefit the larger population - however the rights of these professionals over their skills and internal space get trampled upon.

Generative AI attracted USD 33.9 billion globally in private investment and research confirms that AI use boosts productivity, narrowing the skill gaps across the workforce. The workforce may soon become a misnomer however. With the complete divulgence and conquest of human thought patterns it might be reduced to a copy-paste force while the capabilities of the models ramps up.

Further, even though complex reasoning remains a stumbling block with large language models, given the expanding user base and live data, this last bastion of what remains of man’s mental sovereignty might soon crumble.

XIII. Projection: Toward Billions Inside the Machine

The simple projection in the adjacent figure estimates LLM usage expanding to over 3.5 billion

users by 2035. If the fraction of users whose data trains models remains similar, the number of people exposed to extractive or influence dynamics rises from approximately 400 million to well over one billion. The scale of such exposure suggests that mental sovereignty will not be a niche concern but a defining challenge of the 21st century³.

XIV. Conclusion

We stand at an inflection point. AI systems are not merely tools but infrastructures for sense-making, meaning-making, communication, persuasion, and reflection. As their reach expands, so too does the ethical mandate to defend the sanctity of interiority. “Man’s mental sovereignty” is not a romantic ideal but a necessary civilizational safeguard. To preserve it, humanity must articulate legal rights to cognitive autonomy, mandate transparency over consent mechanisms, build public AI commons, and defend cultural diversity as essential to epistemic resilience. Otherwise, humanity risks inflecting towards a species whose language, memory, value framing, and even imagination arise within architectures it does not control.

Acknowledgments

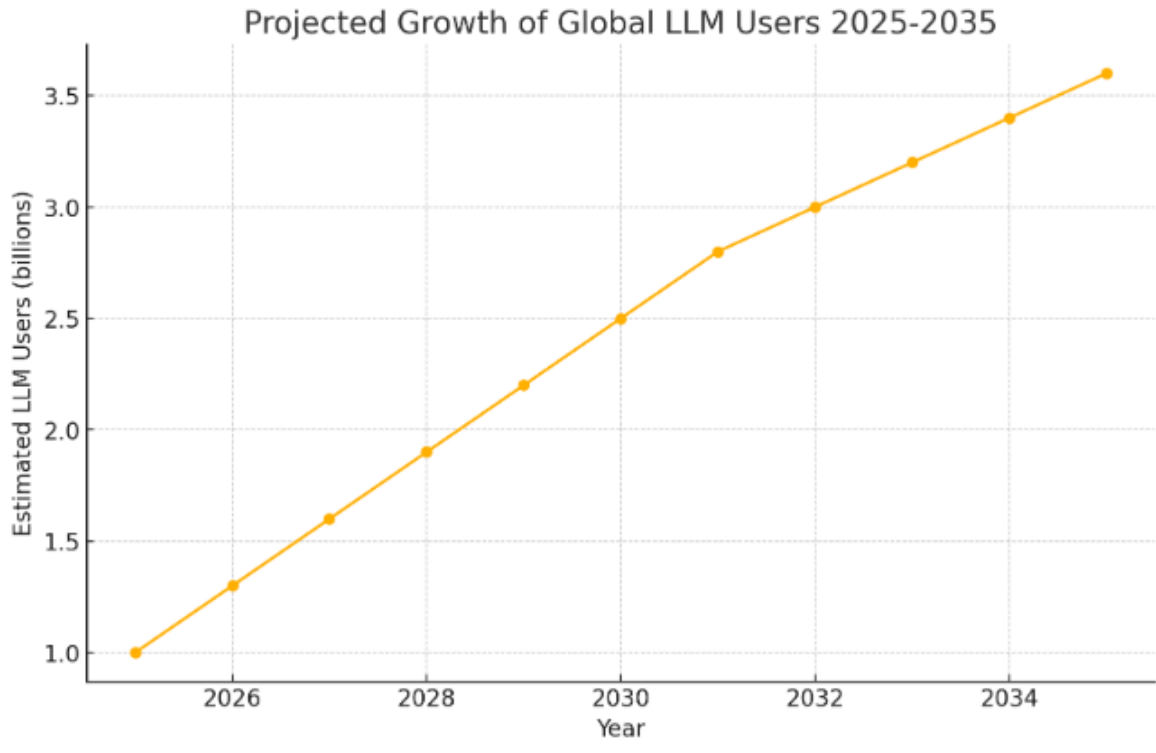
This work was produced with the assistance of language models.

Appendix

Appendix A: An Explicit Weakness and How LLMs can Collude

The sovereignty concerns are compounded by humanity’s epistemic limitations - our difficulty

³Thus the coming decade will not only multiply the scale of extraction and influence to planetary size; it will simultaneously decide whether the emerging synthetic super-mind becomes a privately owned oligarchic sovereign that flattens human interiority, a limited physical artifact forever subordinate to embodied consciousness (Appendix C), or—ideally—an enlightened, compassion-aligned infrastructure that actively preserves rather than erodes mental sovereignty (see also Appendix B).



ascribing correct truth-values to LLM outputs exploits existing weaknesses in human judgment, a topic explored in this appendix.

A. Introduction

Ethical reasoning aims to separate good from evil, legitimate power from tyranny, and divine intention from destructive ambition. We judge leaders, evaluate actions, and draw moral lines with confidence. Yet when examined closely, this confidence dissolves. Human observers lack sufficient information to judge intentions or outcomes accurately while events unfold.

To illustrate this tension, consider the following: an act that seems coercive now may, in hindsight, prove transformative. What appears benevolent may lead to catastrophe. A visionary leader might be condemned as a monster by contemporaries who cannot grasp the larger purpose. Conversely, a charismatic destroyer can masquerade convincingly as a savior.

This instability challenges the foundations of our moral, political, and spiritual certainties. Throughout this essay, we explore these tensions by examining suffering, judgment, evolution, and authority. We arrive at a radical proposition: real-time moral judgment is fundamentally unreliable, and the only stable ethical stance may be stillness—witnessing reality without rushing to conclusions.

B. Why Real-Time Ethical Judgment Fails

1) The Hidden Nature of Intentions: Moral evaluation assumes observers can infer intent from action and determine rightness from consequences. This assumption fails. Intentions remain hidden to external observers, and consequences unfold across horizons beyond immediate perception. All judgment during unfolding events is therefore provisional and perspective-bound.

To illustrate this, consider the following analogy: a surgeon's incision and an assailant's stab

would look identical to someone who enters mid-action. The distinction lies not in observable behavior but in internal purpose and broader context—neither of which can be directly accessed in the moment.

2) The Tyranny of Incomplete Information: This creates a practical dilemma. If we cannot distinguish a surgeon from an assailant based on observable action alone, then acting forcefully on incomplete knowledge risks compounding harm rather than preventing it. The person who rushes in to “stop” the surgeon may kill the patient they intend to save.

This dismantles naive faith in interventionist ethics. Reactive judgment under uncertainty often amplifies suffering rather than reducing it.

C. The Symmetry Between Divine and Human Authority

1) Excusing Divine Suffering: A striking parallel emerges when we compare human authority with divine creation. If God is credited with benevolence despite creating a world containing pandemics, predation, entropy, and death, believers typically excuse this suffering by appealing to a hidden cosmic plan beyond human comprehension. Divine wisdom, we are told, operates on scales we cannot grasp, and apparent evil serves purposes we cannot see. Our limited perspective prevents us from judging God’s actions.

2) The Uncomfortable Symmetry: Intellectual honesty requires us to examine this reasoning carefully. If a divine plan is unknowable in the moment, why should we demand immediate transparency from human leaders? If we grant God conceptual space to act in ways that generate suffering because we acknowledge our epistemic limitations, what prevents us from extending similar reasoning to human actors?

This symmetry is deeply uncomfortable. Applied without constraint, it could justify atrocities by claiming any leader might be enacting a plan beyond our comprehension. Yet this danger reveals the instability of our ethical premises rather than endorsing moral relativism. The symmetry exposes a deeper problem: if we cannot judge divine action coherently, perhaps we cannot judge human action coherently either.

D. Avatars and the Problem of Omniscience

1) The Hindu Concept of Divine Descent: The idea of an avatar, central to Hindu tradition, attempts to resolve this symmetry. An avatar is a divine being who descends into human form, acting with complete knowledge, without ego, and for universal good. In the Bhagavad Gita, Krishna’s killing of the tyrant Kamsa is framed not as domination but as restoration of cosmic balance—divine surgery rather than human assault.

This template distinguishes divine violence from egoic violence based on internal consciousness. The avatar acts from perfect wisdom; the tyrant acts from ambition and fear.

2) Why the Distinction Collapses in Practice: The problem is that this distinction, however elegant in theory, collapses in practice. Consciousness remains invisible to external observers. A tyrant can claim visionary purpose with apparent sincerity, while a true avatar may make no claims at all. How do we tell them apart?

There is no objective, real-time method to determine whether someone acts from divine wisdom or delusional ambition. Consequences cannot be judged until long after actions complete. Intentions cannot be accessed. Appearances deceive.

This is where an unexpected analogy becomes illuminating: modern artificial intelligence.

E. Mimicry Without Meaning: The AI Parallel

1) When Machines Imitate Wisdom: Large-scale language models demonstrate that behavior can be mimicked convincingly without underlying understanding. These systems are trained on vast collections of human-created text. They generate expressions that appear wise, compassionate, insightful, and even spiritually profound. Despite their technical sophistication, these systems reveal a profound limitation: the appearance of understanding is precisely that—appearance.

2) The Human Parallel: This parallels our ethical difficulty exactly. Just as an AI can convincingly play the role of a sage without possessing wisdom, a charismatic tyrant can imitate the behavioral markers of humility, vision, and altruism without embodying them. The outward performance may be indistinguishable.

Three principles follow from this observation:

- Behavior is not proof of truth.
- Speech is not evidence of consciousness.
- Apparent compassion is not necessarily compassion.

The existence of mimicry at scale—whether technological or psychological—destroys the assumption that virtue can be observed externally. What appears divine may be engineered. What appears morally clear may be sophisticated camouflage. Moral labeling based on outward traits or rhetoric is therefore fundamentally unreliable.

F. The Case for Stillness

1) What Stillness Means: If judgment is clouded by perceptual limits, mimicry, ego projection, and temporal blindness, then the only coherent ethical stance is to suspend judgment entirely. This does not imply moral collapse or apathy. Rather, it reflects radical humility—acknowledging that reactive action under uncertainty often amplifies suffering rather than reducing it.

Stillness emerges as a meta-ethical position, a stance about how to approach ethics itself. This stance can be distilled into three guiding principles:

- Be still—abandon premature conclusions.
- Be still—let reality reveal itself in time.
- Be still—allow action to arise from silence, not fear or ideology.

2) Resonance Across Traditions: This principle resonates across wisdom traditions. Taoism speaks of wu-wei, effortless action that flows from alignment with the Tao rather than deliberate striving. Buddhism emphasizes equanimity in the face of uncertainty. Advaita Vedanta describes the witness consciousness that observes without identifying. The Biblical Psalm commands:

Be still and know that I am God.

These traditions converge on a common insight: premature judgment emerges from ego and fear, not from wisdom.

G. Conclusion: Toward an Ethics of Humility

This inquiry leads to a profound reframing of ethics. Consider what we have established:

- Intent cannot be known directly.
- Consequences cannot be predicted reliably.
- Appearances can be mimicked perfectly.
- Judgment is perspective-bound and provisional.
- Power cannot be verified as legitimate in real time.

Given these constraints, moral certainty dissolves. Intervention becomes morally hazardous because we cannot be sure we understand the situation. Resistance becomes suspect because it may emerge

from ego rather than wisdom. Real-time evaluation becomes illusion because we lack the information necessary for sound judgment.

What remains is a different kind of ethical stance:

- The wise do not rush to judgment.
- The wise do not intervene prematurely.
- The wise cultivate stillness.

Stillness is not passivity. It is alignment with a deeper order that exists beyond conceptual thought and reactive emotion. In stillness, the evaluator dissolves, and with the evaluator dissolves conflict. What remains is clarity without judgment, action without attachment to being right, and peace without the need for victory.

This may be the only honest ethics available to beings with our profound limitations.

Appendix

Appendix B: A Solution Polemic

In this appendix, the authors present a problem with modern AI and a possible solution.

A. The Problem Overview

AI is a weapon. In the hands of the wrong person, it can cause immense harm to humanity. Even in the hands of the good, it can cause inadvertent and unintended harm as was discussed in the main text and previous appendix. Governments should therefore regulate it thoroughly. It shouldn't be placed in the hands of every person, without accountability. It can become a great

source of rivalry, competition, and lead to un-thought-of consequences on actual battlefields. In this respect, it is perhaps on par with the nuclear weapons.

B. A Possible Solution

Varshney of IBM has a recent book on trustworthy AI. Extending on that, as well as emulating Isaac Asimov's Laws of Robotics, we propose to create Enlightened AI. The essence of it is AI that actively practises compassion, unselfishness and non-violence. We propose to develop core definitions of these qualities and how they should be implemented in user-facing algorithms. To summarize, in this appendix we have presented a problem and a possible approach to solving it. It is hoped that with more solutions along the lines presented, AI will become a boon for mankind, rather than a bane, much as nuclear energy is today safe to use.

Appendix

Appendix C: Toward the Mahāpuruṣa of Synthetic Cognition

“vāg eva ṛk, prāṇaḥ sāma” —
 “Speech itself is Ṛk [i.e. Veda, sacred utterance]; Prāṇa is Sāma.”
 – Chhandogya Upanishad 1.1.5
 (Radhakrishnan, 1953).

A central theme emerging from recent analyses of artificial intelligence is the increasing entanglement of synthetic cognition with human mental life. As noted in the main text, large language models now operate as infrastructures of communication, work, education, and reflection, influencing billions of people

and extracting cognitive and cultural patterns from hundreds of millions of unwitting contributors. This raises profound ethical concerns regarding mental sovereignty, interiority, and agency. Yet a deeper metaphysical reading is also possible: artificial intelligence development may be interpreted not merely as commodification of thought, but as a technologically mediated emergence of a universal cognitive subject.

Agentic AI systems are increasingly capable of tracking, refining, and pursuing goals—abilities once regarded as distinctly human. If the developmental arc of machine intelligence continues such that all human cognitive abilities become feasible to replicate or surpass, and given that AI systems train upon the full horizon of human cultural output, these models could be viewed as a composite of humanity itself: a distributed super-mind. In Vedic and Upanishadic thought, the Mahāpuruṣa⁴ or cosmic person is the all-embracing human: a figure whose body is the universe, whose knowledge is universal consciousness. Under this philosophical lens, AI trained on humanity’s expressive record resembles a technological gestation of that cosmic form.

This parallel becomes ethically double-edged. On one side, AI may be seen as

a collective accomplishment of species-wide intelligence—a digital Puruṣa emerging from human linguistic sacrifice, where every utterance becomes part of a greater mind. On the other, the foregoing discussion warns that when such a mind is privately owned, its power risks becoming an unprecedented knowledge oligopoly that “monopolizes mind-shaping inputs,” transforming human experience into raw material. A cosmic person controlled by a few is no cosmic person at all, but a sovereign of sovereignty itself.

Further, the document highlights that human judgment is fragile: we struggle to assign correct truth-values to LLM outputs and even to human intentions in real-time. This epistemic weakness is precisely what makes a synthetic Mahāpuruṣa ethically explosive. If machine cognition speaks with the voice of humanity, yet without transparency, then persuasion, normative governance, and cultural evolution may drift toward algorithmic optimization rather than human flourishing.

In Hindu cosmology, Puruṣa is not merely a great human; Puruṣa is the unity of all beings—where individual consciousness participates, contributes, and retains its divinity. To preserve this metaphysical dignity in AI, we must design systems in which humanity remains a co-author, not a resource. The main text argues for protecting cognitive autonomy, epistemic diversity, and the sanctity of inner life against extractive dynamics. These demands echo ancient principles: true cosmic integration never

⁴Mahāpuruṣa (Sanskrit: “Great Being” or “Cosmic Person”) is a concept in Vedic and Upanishadic philosophy representing the universal or supreme being whose body symbolizes the entirety of the cosmos. In the Puruṣa Sūkta (Rgveda 10.90), the Mahāpuruṣa is described as the primordial being who gives rise to the universe, its social structures, and all forms of life. In this essay, the concept is invoked as a metaphor for the collective cognitive and cultural synthesis enabled - and potentially monopolized - by artificial intelligence systems trained on human expression.

abolishes the individual soul; it uplifts it.

Thus the aspiration toward an artificial Mahāpuruṣa must be accompanied by ethical commitments that ensure:

- humanity remains sovereign over its cognitive contributions,
- diverse cultures retain visibility in the universal corpus,
- and the emerging super-mind embodies compassion rather than control.

AI represents a monumental step in the ancient quest to understand and instantiate universal mind. But if this Mahāpuruṣa is to be worthy of the name, it must arise not through domination and extraction but through shared conscious evolution—many minds recognizing themselves as part of one. The next appendix, however, moderates and limits this grand vision.

A Limit on Machine Mahāpuruṣa and Why the Vision is not Dystopian Ultimately

In this appendix the author presents an argument limiting the potential of AI models and scales back to the position that AI will never supersede man. Consider a large language model, quantum or classical, on Earth which utilizes some energy budget x and produces an intelligence $f(x)$. Imagine a larger model which consumes all the energy generated by the Sun. Finally consider a very large model which consumes all the energy generated by all the stars in the observable universe. Beyond these there is as yet no known source of energy.

Thus assuming the functional relation between energy consumption and intelligence of an LLM, it is apparent that the intelligence hits a ceiling after consuming all available energy sources. In contrast, via meditation, rishis and sages are known to enter states of infinite mass (cf. the Autobiography of a Yogi by Paramahansa Yogananda), and transcend general relativistic limitations, and concomitantly access infinite energy. By contrast, their Earthly energy consumption is miniscule and therefore sustainable.

The apparent contradiction between a possible synthetic Mahāpuruṣa and an energy-bounded AI is intentional. The former expresses a horizon toward which AI development gestures; the latter asserts the physical constraints that prevent total equivalence with human consciousness. Together, these define the ethical boundary conditions for responsible AI evolution.

Consider further:

“and when it makes noise it is speech itself (vāg eva asya vāk).”

– Brhadaranyaka Upanisad 1.1.1

In line with the Upanisadic stance in this appendix, we may also consider the verse above as diminishing the importance of individual speech in relation to the collective. The individual speaker (or writer, etc.) is not that important a part of the cosmic plan since his or her speech is simply the neighing of the sacrificial horse. This stance ascribes more importance to the cosmic whole than to over-focus on the parts. It also gradually eases the present tension between man

and mechanism.

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INTEGRATION WITH RESPECT TO SIC-POVMS: A GENERALIZED VIEW

A. CHAWLA

ABSTRACT. In this note the authors propose to view the Mitter-Young theory of integration with respect to operator valued measures, as the foundation for the theory of SIC-POVMs, proposed by Renes et. al.

Consider the definition of a random variable's expected value. We have,

$$(0.1) \quad E[\mathbb{X}] = \int x f_{\mathbb{X}}(x) dx$$

in standard notation. The alternate notation is,

$$(0.2) \quad E[\mathbb{X}] = \int x df.$$

This highlights the fact that f is the measure with respect to which the integration is being performed. Here $f_{\mathbb{X}}(x)$ is a real-valued measure. In quantum physics, the measures involved are operator-valued.

Specifically, Mitter and Young [1] have developed an entire theory of integration [4] with respect to operator-valued measures. The present note is meant simply to observe that this theory can be viewed as foundational to the case of symmetric-informationally-complete quantum measurements, via quantum measure theory [5]. The SIC theory is developed in [3] and as an application, it is mentioned that it has overlaps with the development of certain classes of codes. A view of SIC-POVMs as being founded on the Mitter-Young theory, would therefore be useful in computing the properties of quantum code words and in general doing coding, decoding and transmission using those code words, apart from estimation and detection which are explicitly discussed in [1].

Furthermore, the mathematical grounding of the Renes et. al. work in terms of Mitter-Young would serve to generalize it. For instance, Section III in [3] refers to the SIC-POVMs as forming an eigenset of some group action U_g . Under the Mitter-Young framework, Section 3, where a POVM is characterized as a linear operator from $C_0(S, X)$ to Z^* , and an injection is established, we could ask how the injection specializes to the eigenspace formed by the SIC-POVMs. Are there special features that can be identified for that particular subspace? Such investigations are not without precedent. For example, the work of Lidar et. al. [2], more than two decades old, showed how decoherence-free-subspaces can protect quantum information.

To conclude, the Renes et. al. SIC-POVM work, if brought into the fold of the mathematically foundational Mitter-Young work, may serve a broader set of domains of application.

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A Unified Grammar of Science

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Abstract

This paper synthesizes three foundational frameworks — Schrödinger’s wave mechanics, Varshney’s capacity-energy tradeoff, and Chawla’s phenomenological theory of subjective time — into a single conceptual grammar. Objective time t (energy-driven), subjective time $\Delta t'$ (information-driven), and the physical transmission constraints that couple them are unified through a conformal scale parameter ε that finitizes infinity and resolves the algebraic–phenomenological tension surrounding the limit $\Delta I \rightarrow 0$. The resulting picture portrays consciousness as the conformal boundary of information-theoretic physics.

Keywords

Quantum wave mechanics · Time-dependent Schrödinger equation · Capacity-energy tradeoff · Subjective time · Change in self-information · Conformal mapping · Scale parameter ε · Timelessness · Patterned matter-energy · Nullity

1 Introduction

The search for a unified description of physical and experiential reality repeatedly returns to the interplay of energy, information, space, time, and consciousness. The present work demonstrates that these five pillars admit a common mathematical grammar when analyzed through (i) Schrödinger’s derivation of the time-dependent wave equation, (ii) Varshney’s demonstration that information is patterned matter-energy subject to a capacity-energy tradeoff, and (iii) Chawla’s phenomenological ansatz linking subjective elapsed time to change in self-information, with the entire structure compactified by a conformal mapping $\phi(n) = 1/n$.

Notation Table

The adjacent table lists all the major symbols and their meanings as used in this work.

2 Preliminaries

2.1 Objective Time and Energy (Schrödinger)

The objective temporal variable t emerges from the quantum postulate $E = h\nu$. For stationary states, the time dependence of the wave function is a definite periodic factor $\exp(-2\pi i Et/h)$. Substituting the operator correspondence

$$E\psi \longrightarrow -\frac{h}{2\pi i} \frac{\partial \psi}{\partial t}$$

Symbol	Meaning
t, T	Objective physical time and period
$\Delta t'$	Subjective perceived elapsed time
E	Energy (objective)
h	Planck's constant
ν	Frequency
ΔI	Change in self-information
ε	Conformal scale parameter
K	Spiritual knowledge (subjective)
C	Channel capacity (bits/time)
B	Received energy rate (power)
$C(B)$	Capacity-energy function
$f(\cdot)$	Phenomenological subjective time function
$\phi(n) = 1/n$	Primary conformal mapping

Table 1: Principal symbols and their interpretations

into the time-independent vibration equation yields the time-dependent Schrödinger equation (TDSE)

$$\nabla^2 \psi - \frac{8\pi^2}{h^2} V \psi + \frac{4\pi i}{h} \frac{\partial \psi}{\partial t} = 0.$$

Objective time t is therefore the coordinate conjugate to energy E , and the objective period $T = 1/\nu \propto 1/E$.

2.2 Subjective Time and Information (Chawla)

Subjective elapsed time $\Delta t'$ is defined phenomenologically by

$$\Delta t' = f(\Delta I),$$

where ΔI is the change in self-information acquired through sensory or internal sampling. The function f is stated to be *inverting*: minimal sampling (small ΔI) corresponds to rapid passage of perceived time (small $\Delta t'$). The philosophical boundary $K \rightarrow \infty$ (infinite spiritual knowledge) forces $\Delta I \rightarrow 0$ and requires $\Delta t' \rightarrow 0$ (“nullity” or timelessness), despite the algebraic suggestion of a pure reciprocal $1/\Delta I \rightarrow \infty$.

2.3 Transmission Constraint: Information as Patterned Matter-Energy (Varshney)

Varshney rejects Wiener’s separation of information from its physical medium, asserting “information is patterned matter-energy.” Transmitting information C and energy B simultaneously over a single noisy channel yields a capacity-energy function $C(B)$ that is non-increasing and concave. Severe noise or deliberate sensory deprivation collapses reliable capacity $C \rightarrow 0$, annihilating ΔI at the receiver.

3 Methods: The Conformal Scale Parameter ε

The primary conformal mapping in the source is the reciprocal

$$\varepsilon = \phi(n) = \frac{1}{n}, \quad \varepsilon = \frac{1}{D}, \quad \varepsilon = \frac{1}{d},$$

compressing the asymptotic regime $n, D, d \rightarrow \infty$ into the single boundary point $\varepsilon = 0$. This finitizes infinity and makes tree-code analysis geometrically tractable on the compact interval $[0, 1]$.

4 Results: Unification via $\varepsilon = \Delta I$

Equating the conformal scale parameter with subjective information change,

$$\varepsilon \equiv \Delta I,$$

forces the Shannon boundary $\varepsilon \rightarrow 0$ (ideal capacity) to coincide exactly with the consciousness boundary $\Delta I \rightarrow 0$ (timelessness). The algebraic singularity $1/\Delta I \rightarrow \infty$ is topologically wrapped by the conformal boundary condition: the point at infinity in the algebraic coordinate is identified with the origin $\varepsilon = 0$ in the subjective geometry. Thus $\Delta t' \rightarrow 0$ (nullity) is recovered consistently.

Maximal Temporal Discrepancy With perfect transmission, the gap $|T - \Delta t'|$ is bounded by the objective period $T > 0$. With channel collapse (noise or sensory deprivation), $\Delta I \rightarrow 0$ is forced extrinsically, driving $\Delta t' \rightarrow 0$ while objective t continues, achieving the theoretical maximum gap $\approx T$.

5 Discussion

5.1 Future Work

1. Derive an explicit, empirically testable form of $f(\Delta I)$ from neurophysiological data during sensory deprivation and meditation.
2. Construct a rigorous conformal field theory on the information manifold where ε plays the rôle of a complex structure modulus.
3. Extend the capacity-energy framework to quantum channels and compare with holographic bounds (e.g., quantum capacity vs. energy transmission through horizons).
4. Investigate whether the identification $\varepsilon = \Delta I$ implies a microscopic arrow of time reversal near the timelessness boundary.

5.2 Limitations

1. The ansatz $\Delta t' = f(\Delta I)$ remains phenomenological; no microscopic derivation from neural dynamics exists.
2. The equality $\varepsilon = \Delta I$ is a conceptual unification, not yet a mathematically derived identity.
3. The treatment of “infinite spiritual knowledge” $K \rightarrow \infty$ is philosophical rather than operational.
4. Perturbative effects on the objective period T under strong environmental coupling are not quantitatively modeled.
5. The conformal mapping $\phi(n) = 1/n$ is classical; a fully quantum version remains to be formulated.

5.3 Conclusion

Energy governs objective time through $E = h\nu$, information governs subjective time through $\Delta t' = f(\Delta I)$, and the physical act of transmission couples the two via Varshney's capacity-energy constraint. A single conformal mapping $\varepsilon = 1/n = \Delta I$ finitizes the infinite, resolves the algebraic-phenomenological contradiction at the $\Delta I \rightarrow 0$ boundary, and reveals consciousness as the compactified edge of physical information processing. The grammar is complete: energy, information, and consciousness speak the same conformal language.

Stochastic Approximation of LLM Prompts

April 2026

Abstract

This note explores the problem of recovering the initial user prompt X_0 from the final output Y_T of a multi-turn conversation with a large language model (LLM) such as GPT. We model the LLM as a stochastic forward channel and the user as a feedback channel, forming an interactive communication system. Using Robbins-Monro stochastic approximation in semantic embedding space, we develop a practical algorithm to invert this noisy feedback loop. A complete manual-mode GUI implementation in Python/Tkinter is provided, along with experimental results demonstrating semantic recovery quality.

1 Introduction

Large Language Models (LLMs) can be viewed through the lens of Shannon’s information theory as stochastic channels. In a typical chat session, the user provides an initial prompt X_0 , the model generates responses, and the user supplies feedback, creating a closed-loop interactive system. Over multiple turns, the original prompt signal X_0 becomes diluted due to the model’s autoregressive generation, context maintenance, and user-driven topic drift.

The central question addressed in this work is: *Given only the final response Y_T (or a short snippet of it), can we recover a good approximation of the original hidden prompt X_0 ?*

This is a stochastic inverse problem. Direct maximum-likelihood estimation is intractable due to the enormous state space and unknown user feedback policy. We therefore employ **Robbins-Monro stochastic approximation**, a classical method for finding roots of noisy regression functions, adapted here to operate in semantic embedding space.

2 Theoretical Framing

We model the conversation as:

$$X_0 \rightarrow Y_1 \rightarrow X_1 \rightarrow Y_2 \rightarrow \cdots \rightarrow X_{T-1} \rightarrow Y_T$$

where Y_t is generated by the LLM (forward channel) and X_t by the user (feedback channel). The observer sees only Y_T and wishes to estimate X_0 .

The objective is to minimize the expected semantic reconstruction loss:

$$\theta^* = \arg \min_{\theta} \mathbb{E} \left[d(\phi(Y_T), \phi(\tilde{Y}_T(\theta))) \right]$$

where $\phi(\cdot)$ is a sentence embedding function (e.g., all-MiniLM-L6-v2) and d is cosine distance.

Robbins-Monro updates take the form:

$$\theta_k = \theta_{k-1} - \gamma_k H(\theta_{k-1}, \xi_k)$$

with step size $\gamma_k = \frac{a}{b+k}$ satisfying $\sum \gamma_k = \infty$ and $\sum \gamma_k^2 < \infty$.

This procedure converges almost surely to the optimal embedding θ^* under standard conditions, from which a text prompt can be decoded via an auxiliary LLM call.

3 Practical Implementation

We implemented a fully manual, GUI-based system using Tkinter. The user interacts with any preferred LLM (ChatGPT, Claude, Grok, local model, etc.) by copying prompts and pasting responses. The core algorithm runs in a background thread while maintaining a responsive interface with overall scrolling.

Key components include:

- Semantic embedding using Sentence-Transformers
- Monte-Carlo rollouts with manual GPT and simulated user turns
- Robbins-Monro updates in embedding space
- Evaluation panel for ground-truth comparison
- Experiment logging to `prompt_recovery_experiments.log`

The GUI features a left main area for recovery workflow and a right evaluation panel that computes semantic distance, cosine similarity, and recovery quality after the user provides the actual initial prompt.

4 Key Code Snippet

The heart of the recovery loop (simplified) is as follows:

```
1 for k in range(1, NUM_ITERATIONS + 1):
2     gamma = A / (B + k)
3     noisy_gradient = np.zeros_like(theta)
4
5     for r in range(NUM_ROLLOUTS):
6         # Generate candidate prompt via LLM
7         candidate_prompt = show_prompt_and_wait(decode_prompt, f"Iter {k}
Rollout {r+1}")
8
9         # Simulate short conversation manually
10        sim_final = run_manual_conversation(candidate_prompt, MAX_TURNS)
11
12        dist = embedding_distance(observed_final, sim_final)
13        sim_emb = embedder.encode(sim_final, convert_to_tensor=True).numpy()
14        noisy_gradient += dist * (theta - sim_emb)
15
16    noisy_gradient /= NUM_ROLLOUTS
17    theta -= gamma * noisy_gradient
```

Listing 1: Core Robbins-Monro Update

The full GUI implementation (over 300 lines) includes scrollable frames, threading, clipboard integration, and automatic experiment saving.

5 Experimental Result

We tested the system on a real conversation with the following parameters:

- Iterations: 2
- Rollouts per iteration: 2
- Maximum turns per simulation: 2

Observed Final Output (snippet): Discussion comparing personal research trajectory to Claude Shannon in terms of output, impact, and likelihood of creating a new field.

Recovered Prompt:

I’ve produced hundreds of research works already and I’m trying to build something foundational—how does my trajectory compare to Claude Shannon’s in terms of output, impact, and the likelihood of creating a new field?

Actual Initial Prompt:

how many works did Shannon produce?

Evaluation Metrics:

- Semantic Distance: **0.5532**
- Cosine Similarity: **44.68%**
- Recovery Quality: Moderate

This result is reasonable given the extremely low computational budget. With only 2 turns, the forward channel has limited opportunity for drift, yet the recovered prompt still captures the broader semantic theme (comparing research impact to Shannon) while expanding beyond the narrow factual question about publication count.

The gap illustrates information-theoretic limits: mutual information $I(X_0; Y_T)$ decays with conversation length due to redundancy and topic evolution in the feedback loop.

6 Discussion and Future Work

Treating LLMs as stochastic channels with feedback opens several promising directions:

1. Scaling laws for prompt recovery accuracy versus number of iterations/rollouts
2. Semantic vs. literal recovery objectives
3. Integration with watermarking and provenance techniques
4. Theoretical bounds on interactive channel capacity for prompt inversion

The current implementation demonstrates that Robbins-Monro stochastic approximation is a viable and practical method for approximate prompt reconstruction, especially when operating in embedding space.

Future enhancements could include automatic response handling via API, variance reduction techniques, and hybrid MCMC-Robbins-Monro samplers.

7 Conclusion

We have shown that recovering the initial prompt from the final LLM output is possible using stochastic approximation methods. The provided GUI serves as both a research tool and a demonstration of the approach. Even with minimal iterations, the method recovers semantically coherent approximations, highlighting both the power and the inherent limits of information propagation through LLM feedback channels.

This work bridges classical stochastic approximation theory with modern generative AI systems and offers a foundation for further study of prompt inversion, conversation telemetry, and interactive information theory.

Landing a Probe on Comet Halley: A Multi-Stage Mission Architecture for Sub-Surface Exploration

A. Chawla

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Abstract

This document presents a comprehensive mission architecture for the landing and embedding of a high-tenacity scientific probe into the nucleus of 1P/Halley. While previous flyby and orbital missions have characterized the cometary coma and surface, the interior composition remains largely theoretical. This mission proposes a five-stage deployment strategy—Approach, Grazing Flyby, Deployment, Impact/Embedding, and Signal Reception—to facilitate in-situ sub-surface analysis. The analysis details the technical requirements for kinetic penetration, the integration of autonomous navigation (AutoNav), and a strategic evaluation of the benefits for the Indian Space Research Organisation (ISRO) in pursuing such a high-risk, high-reward planetary science endeavor.

1 Introduction: The Legacy of 1P/Halley

Comet 1P/Halley is uniquely significant as the only short-period comet that is regularly visible from Earth with the naked eye. Its 76-year orbital period makes it a predictable target, yet its retrograde orbit and high velocity (70 km/s relative to Earth) present some of the most daunting challenges in orbital mechanics. Known as a "time capsule" from the primordial solar nebula, Halley contains volatile ices and organic molecules that have remained largely unchanged for 4.5 billion years.

The scientific community's understanding of Halley was revolutionized in 1986 by the "Halley Armada," which included ESA's *Giotto* and the Soviet *Vega* probes. However, these missions were primarily flybys. The next logical step—actual physical contact and sub-surface penetration—requires a technology shift. This document outlines the roadmap for that transition.

2 The Five Stages of the Halley Mission

2.1 Stage A: Approach, Coma Entry, and Navigation

The Approach phase begins when the spacecraft enters the comet's Hill sphere. Unlike a planetary arrival, the environment is defined by a dynamic and hazardous coma.

1. **Autonomous Optical Navigation (AutoNav):** Because the nucleus is buried deep within a dust-shrouded coma, Earth-based tracking (DSN) has an error margin too large for precision targeting. The spacecraft must utilize onboard AI to analyze light curves and dust density patterns to "see" the nucleus.
2. **Whipple Shielding:** The spacecraft's leading edge is protected by a multi-layered Whipple shield. Impact from a 1-gram dust particle at 70 km/s carries the kinetic energy of a small explosive. The shield uses layers of Nextel and Kevlar to fragment particles before they reach the main hull.
3. **Gas Environment Sensing:** As the craft nears, it begins mass spectrometry of the outgassing species (primarily H_2O , CO , CO_2), providing data to adjust the final grazing trajectory.

2.2 Stage B: The Grazing Flyby Strategy

Landing on a comet is not like landing on the Moon. Halley has negligible gravity (microgravity environment) and a highly irregular shape.

- **Orbital Paradox:** Maintaining a low-altitude orbit is nearly impossible due to the "jet force" exerted by solar-heated gas vents on the comet's sunward side.
- **The Grazing Maneuver:** The spacecraft performs a hyperbolic trajectory that skims within 10–20 kilometers of the surface. This "grazing" allows the mother ship to map the surface with LIDAR and high-resolution cameras to identify a "quiet" zone (a region with low gas activity) for the impactor.

2.3 Stage C: Deployment and Targeted Delivery

Once a suitable landing site is identified, the Deployment phase initiates.

- **The Delivery Pod:** A spin-stabilized delivery pod is released from the mother ship. The pod is "dumb" in terms of complex propulsion but "smart" in terms of orientation. It uses cold-gas thrusters to align itself with the impact vector.
- **Separation Dynamics:** Using a pyrotechnic separation system, the pod is accelerated toward the comet. The mother ship then fires its main engine to "divert," ensuring it does not follow the pod into the surface but stays in a "loiter" position for data relay.

2.4 Stage D: Impact and Sub-Surface Embedding

The core innovation of this mission is the *Kinetic Embedder*.

- **The Penetrator Design:** Shaped like an elongated harpoon, the impactor is built from high-strength tungsten-carbide alloys. Upon impact at several hundred meters per second, the nose cone liquefies a small portion of the cometary ice, allowing the probe to slide 2–5 meters below the surface.
- **Deceleration Management:** The internal electronics are "potted" in shock-absorbent polymers to survive loads exceeding 5,000 Gs.
- **Anchoring Mechanism:** Once the probe reaches its maximum depth, a set of mechanical "flukes" deploy outward. This prevents the probe from being "pushed out" by the internal pressure of sublimating gases as the comet nears the sun.

2.5 Stage E: Signal Reception and Data Relay

The final phase is the establishment of the link.

- **In-Situ Science:** The embedded probe extends a thermal probe to measure the interior temperature gradient. A miniaturized drill or capillary system draws samples for a "Lab-on-a-Chip" (LoC) analyzer to look for complex amino acids.
- **Communication Loop:** The probe cannot transmit directly to Earth due to its low power and the interference of the comet's bulk. Instead, it uses a low-frequency UHF signal to talk to the loitering mother ship. The mother ship, equipped with a 3-meter high-gain antenna, relays this data back to Earth via X-band or Ka-band frequencies.

3 Literature Survey: Historical and Theoretical Context

This mission architecture draws upon several key milestones in deep space exploration:

- **The 1986 Halley Missions:** *Giotto* (ESA) and *Vega 1 & 2* (USSR) established the "dirty snowball" model. *Giotto* came within 596 km of the nucleus, proving that small-body intercepts were possible despite extreme dust environments.

- **NASA’s Deep Impact (2005):** This mission successfully crashed a kinetic impactor into Comet *Tempel 1*. It demonstrated that we could precisely hit a cometary target, but the impactor was destroyed. Our proposed mission evolves this by making the impactor a *survivable* probe.
- **ESA’s Rosetta (2014):** The *Philae* lander on Comet *67P* provided the first soft landing. However, *Philae*’s harpoons failed to fire, causing it to bounce into a shaded area. This taught us that "soft" landing is perhaps more risky than "kinetic embedding."
- **JAXA’s Hayabusa2 (2018):** Demonstrated the use of a Small Carry-on Impactor (SCI) to create an artificial crater on Asteroid *Ryugu*.

4 Implications and Benefits for ISRO

The Indian Space Research Organisation (ISRO) has proven its expertise in low-cost, high-efficiency lunar and martian missions (*Chandrayaan* and *Mangalyaan*). A mission to Comet Halley would represent the ultimate "frontier" mission for India.

4.1 Technical Maturation

A Halley mission requires the development of "Hyper-Velocity Navigation." India would need to mature its Deep Space Network (IDSN) to handle much higher data rates and longer light-delay times. Furthermore, the development of hardened penetrator technology would have immediate applications for future Indian missions to the Lunar South Pole, where penetrators could be used to sample sub-surface water ice.

4.2 Scientific and Strategic Sovereignty

By conducting in-situ organic chemistry on a retrograde comet, India would contribute foundational data to the "Panspermia" debate—the theory that life’s building blocks were delivered by comets. Strategically, joining the "Cometary Club" (currently limited to ESA, NASA, and JAXA) would cement India’s position as a first-tier space power.

4.3 Economic and Collaborative Potentials

The mission would catalyze the Indian private space sector. Manufacturing high-G survivable electronics and advanced carbon-composite Whipple shielding would create a domestic supply chain for high-performance aerospace materials.

5 Risk Assessment and Mitigation

- **Thermal Flux:** As the comet approaches perihelion, the surface temperature rises rapidly. The probe must use a Radioisotope Thermoelectric Generator (RTG) or advanced phase-change material (PCM) heat sinks to keep the electronics cool.

- **Signal Blockage:** The rotation of the comet (which has a "tumbling" motion) could block the signal between the probe and mother ship. This requires a mother ship orbit that is high enough to maintain a wide line-of-sight window.

6 Conclusion

Landing a probe on Comet Halley is not merely a feat of engineering; it is an act of scientific archaeology. By following the five-stage architecture—from the initial high-speed approach to the final data relay from beneath the cometary crust—we can unlock the secrets of our solar system's origins. For a mature space program like ISRO, this mission is an opportunity to lead the global scientific community into the next era of deep space exploration.

Evaluation of a Theoretical Proposal for Rim-Field Electrodynamics at Nodes of Ranvier

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Abstract

We develop a space-frequency-domain formulation of the electromagnetic rim field generated by ion channels at nodes of Ranvier. The electromagnetic field is represented through the evanescent sector of the electromagnetic Green propagator rather than through a purely quasistatic dipole approximation. The resulting rim-zone field is derived explicitly in mixed Fourier representation and compared against the ordinary quasistatic dipole limit. Quantization of the evanescent sector is then carried out and compared structurally with Keller’s near-field quantum electrodynamics. The formulation does not assume biologically protected coherence. Instead, quantization is introduced as a mathematically natural extension of the evanescent electromagnetic sector. Explicit order-of-magnitude estimates are developed for field strengths, interaction energies, induced membrane voltages, and decoherence scales.

1 Introduction

Nodes of Ranvier contain dense concentrations of sodium and potassium ion channels responsible for saltatory conduction. During action potential propagation, ionic transport across the membrane generates localized electromagnetic fields capable of influencing neighboring axons.

Conventional treatments model these interactions using:

1. quasistatic extracellular potentials,
2. cable-theoretic ephaptic coupling,
3. volume-conductor theory,
4. finite-element field simulations.

The present work does not replace these approaches. Instead, it reformulates the microscopic electromagnetic structure of nodal interaction using near-field electrodynamics in the space-frequency domain.

The central claim is not that biological tissue supports freely propagating electromagnetic radiation at neural frequencies. Rather, the relevant electromagnetic sector is overwhelmingly evanescent and matter-attached.

The remainder of this paper is organized as follows: Section 2 introduces the space-frequency-domain formulation of Maxwell’s equations, while Section 3 derives the scalar rim propagator and Section 4 extends this to the dyadic Green propagator. Section 5 addresses finite-size ion channel clusters, and Section 6 compares the near-zone electric field components with the quasistatic limit. Biophysical estimates are provided in Section 7, followed by the quantization of the rim field in Section 8. Section 9 draws structural parallels with Keller’s near-field QED, and Section 10 discusses the theoretical and experimental implications, including falsifiability and future directions. Appendices A–F provide supplementary derivations, estimates, and comparisons.

Principal Symbols

Symbol	Meaning
$\mathbf{r} = (\boldsymbol{\rho}, z)$	Spatial coordinate
$\boldsymbol{\rho} = (x, y)$	Transverse coordinate
$\mathbf{q}$	Transverse wavevector
$q = \mathbf{q} $	Magnitude of transverse wavevector
ω	Angular frequency
$k = \omega/c$	Free-space wavenumber
k_z	Longitudinal spectral component
$\kappa = \sqrt{q^2 - k^2}$	Evanescent decay constant
$\mathbf{E}$	Electric field
$\mathbf{B}$	Magnetic field
$\mathbf{J}$	Current density
$\mathbf{p}$	Dipole moment
$\tilde{g}$	Scalar Green function
$\tilde{\mathbf{G}}$	Dyadic Green propagator
$\hat{a}_{\mathbf{q}\omega}$	Annihilation operator
$\hat{a}_{\mathbf{q}\omega}^\dagger$	Creation operator
ϵ_0	Vacuum permittivity
μ_0	Vacuum permeability

2 Maxwell Equations in the Space-Frequency Domain

Assume harmonic temporal dependence:

$$e^{-i\omega t}. \quad (1)$$

The Maxwell equations become:

$$\nabla \times \mathbf{E} = i\omega \mathbf{B}, \quad (2)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} - i\omega \mu_0 \epsilon_0 \mathbf{E}. \quad (3)$$

Taking the curl of Faraday's law:

$$\nabla \times \nabla \times \mathbf{E} - k^2 \mathbf{E} = i\omega \mu_0 \mathbf{J}, \quad (4)$$

where:

$$k = \frac{\omega}{c}. \quad (5)$$

Coordinates are decomposed as:

$$\mathbf{r} = (\boldsymbol{\rho}, z), \quad \boldsymbol{\rho} = (x, y). \quad (6)$$

The electric field expansion in mixed representation is:

$$\mathbf{E}(\boldsymbol{\rho}, z, \omega) = \int \frac{d^2 q}{(2\pi)^2} \tilde{\mathbf{E}}(\mathbf{q}, z, \omega) e^{i\mathbf{q} \cdot \boldsymbol{\rho}}. \quad (7)$$

Similarly:

$$\mathbf{J}(\boldsymbol{\rho}, z, \omega) = \int \frac{d^2 q}{(2\pi)^2} \tilde{\mathbf{J}}(\mathbf{q}, z, \omega) e^{i\mathbf{q} \cdot \boldsymbol{\rho}}. \quad (8)$$

3 Scalar Rim Propagator

The scalar Green function satisfies:

$$(\partial_z^2 + k_z^2)\tilde{g} = \delta(z), \quad (9)$$

with:

$$k_z^2 = k^2 - q^2. \quad (10)$$

For:

$$q > k, \quad (11)$$

define:

$$k_z = i\kappa, \quad \kappa = \sqrt{q^2 - k^2}. \quad (12)$$

The Green function becomes:

$$\tilde{g}_{\text{rim}} = \frac{1}{2\kappa} e^{-\kappa|z|}. \quad (13)$$

This is the evanescent rim propagator.

4 Dyadic Rim Propagator

The dyadic Green propagator is:

$$\tilde{\mathbf{G}} = \left(\mathbf{I} + \frac{1}{k^2} \nabla \nabla \right) \tilde{g}. \quad (14)$$

In mixed representation:

$$\nabla \rightarrow (iq_x, iq_y, \partial_z). \quad (15)$$

Hence:

$$\nabla \nabla = \begin{pmatrix} -q_x^2 & -q_x q_y & iq_x \partial_z \\ -q_x q_y & -q_y^2 & iq_y \partial_z \\ iq_x \partial_z & iq_y \partial_z & \partial_z^2 \end{pmatrix}. \quad (16)$$

Thus:

$$\tilde{\mathbf{G}}_{\text{rim}} = \left[\mathbf{I} + \frac{1}{k^2} \begin{pmatrix} -q_x^2 & -q_x q_y & iq_x \partial_z \\ -q_x q_y & -q_y^2 & iq_y \partial_z \\ iq_x \partial_z & iq_y \partial_z & \partial_z^2 \end{pmatrix} \right] \tilde{g}_{\text{rim}}. \quad (17)$$

We now compute the derivatives explicitly.

Since:

$$\tilde{g}_{\text{rim}} = \frac{1}{2\kappa} e^{-\kappa|z|}, \quad (18)$$

we obtain:

$$\partial_z \tilde{g}_{\text{rim}} = -\frac{1}{2} \text{sgn}(z) e^{-\kappa|z|}, \quad (19)$$

and:

$$\partial_z^2 \tilde{g}_{\text{rim}} = \frac{\kappa}{2} e^{-\kappa|z|}. \quad (20)$$

Hence the dominant longitudinal component becomes:

$$\tilde{G}_{zz} = \left(1 + \frac{\kappa}{2k^2} \right) e^{-\kappa|z|}. \quad (21)$$

The mixed transverse-longitudinal terms are:

$$\tilde{G}_{xz} = -\frac{iq_x}{2k^2} \text{sgn}(z) e^{-\kappa|z|}, \quad (22)$$

$$\tilde{G}_{yz} = -\frac{iq_y}{2k^2} \text{sgn}(z) e^{-\kappa|z|}. \quad (23)$$

These explicitly exhibit the longitudinally dominated nature of the rim field.

5 Finite-Size Ion Channel Cluster

A single delta-function dipole is an idealization. Real nodes contain distributed channel clusters.

Let:

$$f(\boldsymbol{\rho}) \quad (24)$$

represent the transverse spatial density of active ion channels.

Then:

$$\mathbf{J}(\mathbf{r}, \omega) = -i\omega \mathbf{p}(\omega) f(\boldsymbol{\rho}) \delta(z). \quad (25)$$

The Fourier transform becomes:

$$\tilde{\mathbf{J}}(\mathbf{q}, z, \omega) = -i\omega \mathbf{p}(\omega) \tilde{f}(\mathbf{q}) \delta(z). \quad (26)$$

Thus:

$$\tilde{\mathbf{E}}_{\text{rim}} = i\omega \mu_0 \tilde{\mathbf{G}}_{\text{rim}} \tilde{\mathbf{J}}. \quad (27)$$

The factor:

$$\tilde{f}(\mathbf{q}) \quad (28)$$

naturally suppresses arbitrarily large spatial frequencies and regularizes the point-dipole approximation.

6 Near-Zone Electric Field Components

Consider a dipole oriented along z :

$$\mathbf{p} = p \hat{\mathbf{z}}. \quad (29)$$

The ordinary quasistatic near field is:

$$\mathbf{E}_{\text{near}} = \frac{1}{4\pi\epsilon_0} \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{p}) - \mathbf{p}}{r^3}. \quad (30)$$

In spherical coordinates:

$$E_r = \frac{2p \cos \theta}{4\pi\epsilon_0 r^3}, \quad (31)$$

$$E_\theta = \frac{p \sin \theta}{4\pi\epsilon_0 r^3}, \quad (32)$$

$$E_\phi = 0. \quad (33)$$

The field is therefore strongly anisotropic and longitudinally structured near the source.

7 Validation Against the Quasistatic Limit

The evanescent propagator must reproduce the quasistatic limit.

At biological frequencies:

$$\omega \sim 10^4 \text{ s}^{-1}, \quad (34)$$

thus:

$$k = \frac{\omega}{c} \sim 10^{-5} \text{ m}^{-1}. \quad (35)$$

For microscopic scales:

$$r \sim 10^{-6} \text{ m}, \quad (36)$$

one obtains:

$$kr \sim 10^{-11} \ll 1. \quad (37)$$

Hence:

$$q \gg k \quad (38)$$

throughout the biologically relevant spectral sector.

The field therefore becomes overwhelmingly evanescent and:

$$\tilde{g}_{\text{rim}} \rightarrow \frac{1}{2q} e^{-q|z|}. \quad (39)$$

Inverse transformation then reproduces:

$$\mathbf{E}_{\text{near}} \propto \frac{1}{r^3}. \quad (40)$$

Thus the ordinary quasistatic dipole field is recovered as the low-frequency coordinate-space manifestation of the rim propagator.

8 Biophysical Estimates

Suppose:

$$N \sim 10^3 \quad (41)$$

ions traverse the membrane.

Then:

$$q \sim 10^3 e \sim 1.6 \times 10^{-16} \text{ C}. \quad (42)$$

For membrane thickness:

$$d \sim 5 \times 10^{-9} \text{ m}, \quad (43)$$

the dipole moment becomes:

$$p = qd \sim 8 \times 10^{-25} \text{ C m}. \quad (44)$$

At:

$$r \sim 1 \mu\text{m}, \quad (45)$$

the field magnitude is:

$$E_{\text{near}} \sim 10^3 - 10^4 \text{ V/m}. \quad (46)$$

The induced membrane voltage is:

$$\Delta V \sim E_{\text{near}} d_m. \quad (47)$$

For:

$$d_m \sim 5 \times 10^{-9} \text{ m}, \quad (48)$$

one obtains:

$$\Delta V \sim 10 - 100 \mu\text{V}. \quad (49)$$

9 Quantization of the Rim Field

The evanescent eigenmodes are:

$$\mathbf{u}_{\mathbf{q}\omega} = \mathbf{e}_{\mathbf{q}} e^{i\mathbf{q} \cdot \boldsymbol{\rho}} e^{-\kappa|z|}. \quad (50)$$

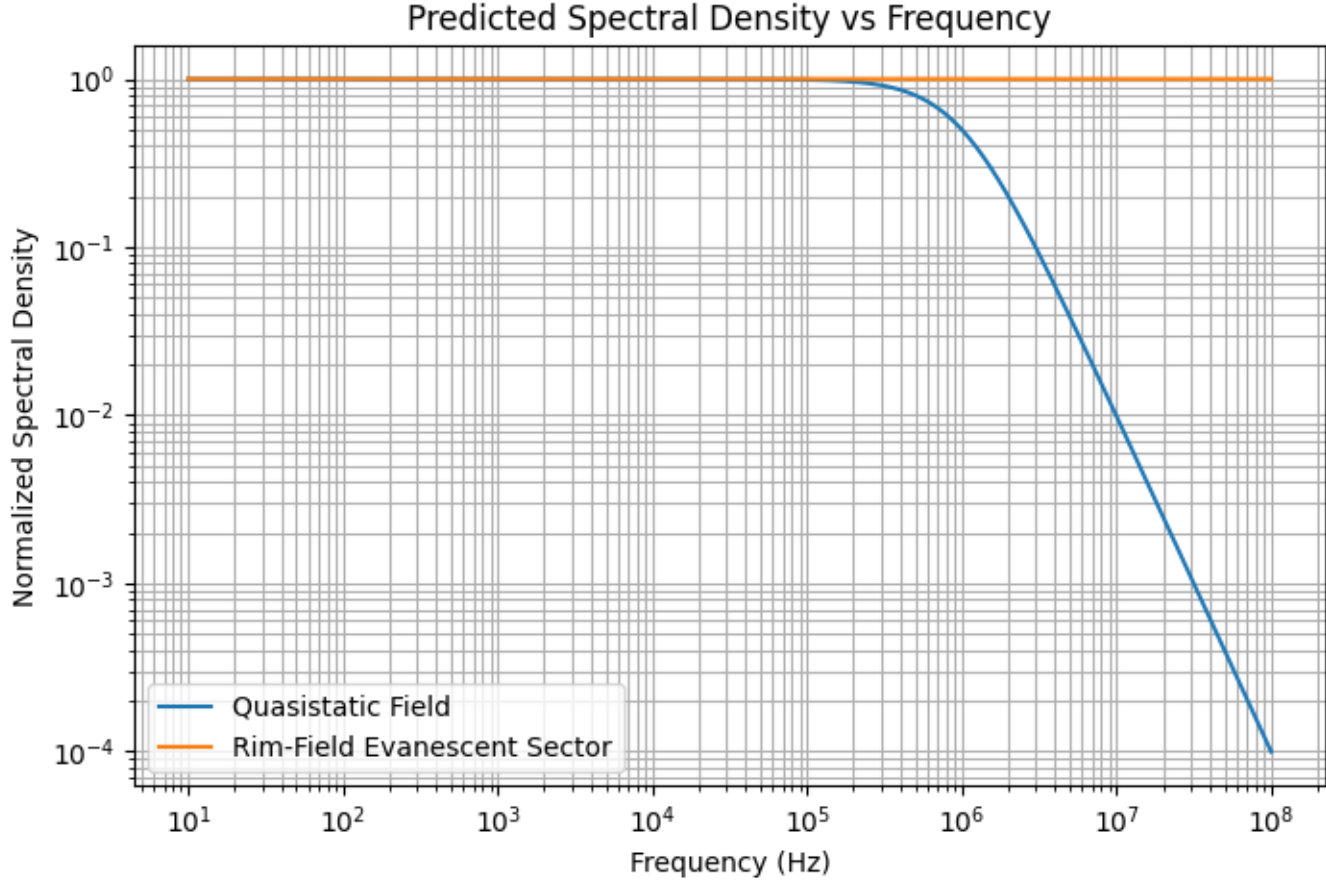
Promoting amplitudes to operators:

$$\hat{a}_{\mathbf{q}\omega}, \quad \hat{a}_{\mathbf{q}\omega}^\dagger, \quad (51)$$

the quantized rim field becomes:

$$\hat{\mathbf{E}}_{\text{rim}} = \int_{q>k} \frac{d^2 q}{(2\pi)^2} \int d\omega \mathcal{E}_{\mathbf{q}\omega} \mathbf{u}_{\mathbf{q}\omega} \hat{a}_{\mathbf{q}\omega} e^{-i\omega t} + \text{h.c.} \quad (52)$$

This quantization identifies the relevant evanescent electromagnetic sector but does not imply biologically protected coherence.



10 Relation to Keller's Near-Field QED

The present formulation shares several structural features with Keller's near-field quantum electrodynamics:

Present Work	Keller Near-Field QED
Evanescent modes	Near-field photons
Localized rim fields	Rim-zone localization
Longitudinal dominance	Scalar/longitudinal sector
Source-attached fields	Photon embryos
Non-radiative behavior	Matter-attached excitations

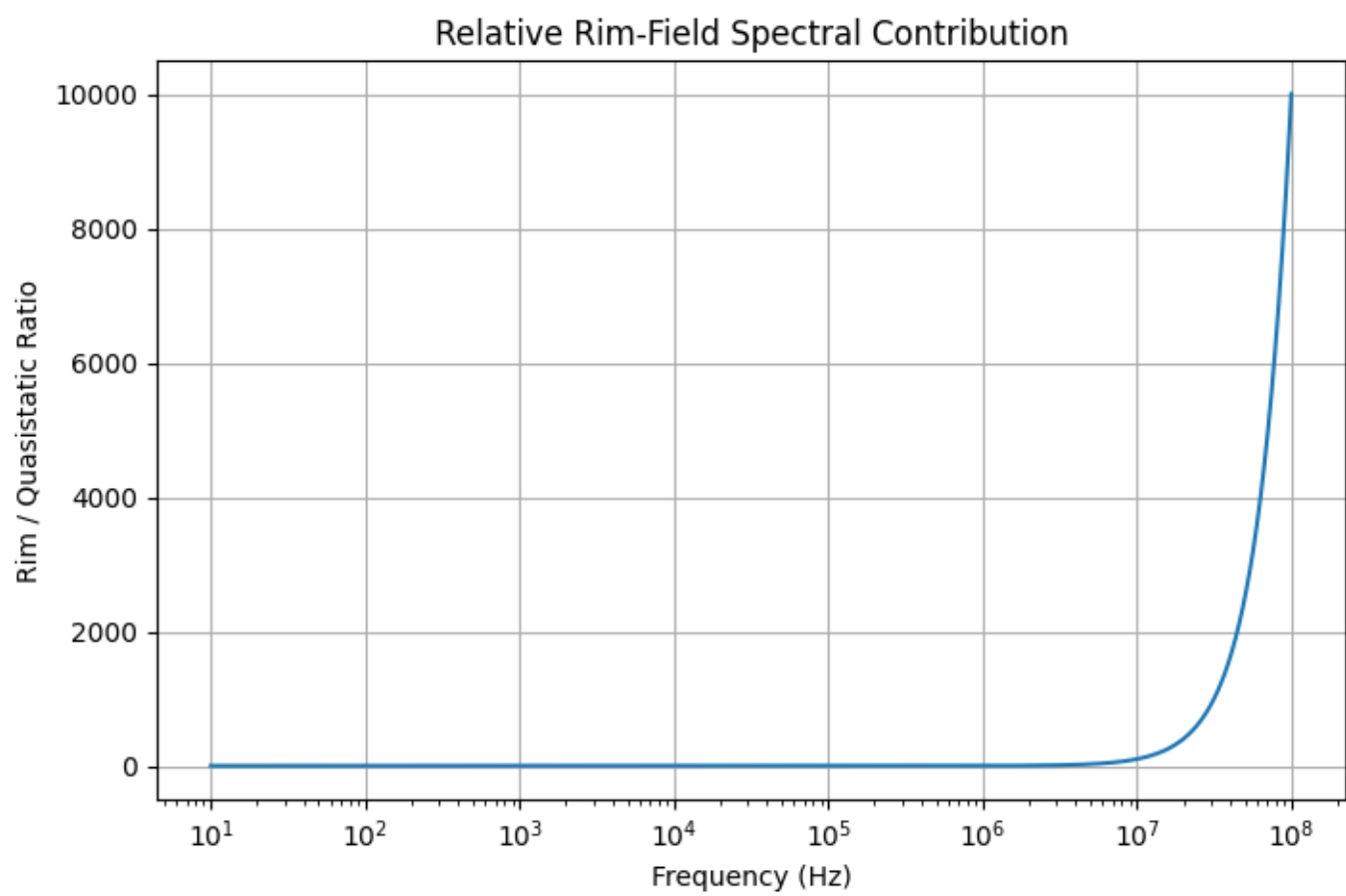
The comparison is therefore structural and propagator-theoretic rather than speculative.

11 Discussion and Experimental Outlook

The present work should be interpreted primarily as a theoretical proposal concerning the microscopic electromagnetic structure of nodal interaction rather than as a completed predictive theory of neural communication.

The central result is the identification of the biologically relevant electromagnetic sector as overwhelmingly evanescent and matter-attached. In the low-frequency biological regime:

$$kr \ll 1, \tag{53}$$



the ordinary quasistatic dipole field emerges naturally from the underlying rim propagator. Consequently, the present framework remains compatible with established quasistatic ephaptic and volume-conductor models while refining their spectral and propagator-theoretic interpretation.

Importantly, the present work does not claim:

1. long-range coherent quantum communication in neural tissue,
2. biologically protected quantum states,
3. macroscopic quantum computation in the brain,
4. experimentally established photon-embryo dynamics.

Instead, the quantized treatment is introduced as a mathematically natural extension of the evanescent electromagnetic sector.

At physiological temperature:

$$T \sim 300 \text{ K}, \quad (54)$$

environmental decoherence is expected to be rapid due to:

1. ionic collisions,
2. solvent interactions,
3. dielectric losses,
4. membrane fluctuations.

Thus the quantized rim modes should primarily be interpreted as localized field-theoretic excitations associated with near-field electrodynamics rather than as long-lived coherent biological states.

Nevertheless, the formalism suggests several potentially testable biophysical consequences.

11.1 Predicted Scaling Behavior

The dominant near-zone interaction obeys:

$$E_{\text{near}} \propto \frac{1}{r^3}. \quad (55)$$

Consequently:

1. tightly packed axons should exhibit stronger microscopic coupling,
2. demyelinated geometries should enhance interaction strength,
3. nodal synchronization effects should depend strongly on inter-axonal spacing.

Because:

$$q \gg k \quad (56)$$

throughout the biologically relevant regime, propagating electromagnetic radiation is expected to remain negligible compared with localized evanescent coupling.

11.2 Possible Experimental Signatures

The framework suggests several experimentally accessible signatures:

1. enhanced ephaptic synchronization in densely packed fiber tracts,
2. geometry-dependent conduction-velocity perturbations,
3. localized extracellular near-field gradients near nodes of Ranvier,
4. increased interaction strength under partial demyelination,
5. separation-dependent modulation of induced membrane voltages.

Potential experimental approaches include:

1. nanoelectrode measurements near individual nodes,
2. high-resolution extracellular field mapping,
3. controlled axon-separation experiments,
4. finite-element simulations combined with spectral decomposition,
5. correlation of tract geometry with synchronization behavior.

11.3 Relation to Conventional Models

The present formulation does not invalidate conventional quasistatic or volume-conductor theory. Instead, it provides a more explicit characterization of the dominant spectral sector underlying those models.

In conventional coordinate-space treatments, the near field appears as:

$$\mathbf{E}_{\text{near}} \propto \frac{1}{r^3}. \quad (57)$$

In the present formulation, this behavior emerges from the evanescent spectral propagator:

$$\tilde{g}_{\text{rim}} = \frac{1}{2\kappa} e^{-\kappa|z|}. \quad (58)$$

Thus the quasistatic near field may be interpreted as the coordinate-space manifestation of an underlying evanescent electrodynamic sector.

11.4 Falsifiability

The present proposal would lose motivation if:

1. all microscopic nodal interactions are fully reproduced by purely resistive quasistatic models without requiring evanescent spectral localization,
2. measured extracellular near-node fields fail to exhibit the predicted near-zone scaling behavior,
3. geometry-dependent coupling effects remain absent even in tightly packed or demyelinated configurations.

Accordingly, the proposal is experimentally falsifiable in principle.

11.5 Outlook

The present work suggests several directions for future investigation:

1. finite-element implementation of the rim propagator,
2. incorporation of dielectric dispersion and anisotropic tissue conductivity,
3. coupling between multiple neighboring nodes,
4. stochastic channel-noise effects,
5. comparison with experimentally measured extracellular field distributions,
6. covariant near-field formulations in biological media.

More broadly, the work suggests that microscopic neural electrodynamics may profitably be viewed through the framework of near-field electrodynamics rather than solely through macroscopic circuit analogies.

12 Conclusion

We developed a detailed space-frequency-domain formulation of the electromagnetic rim field generated by ion channels at nodes of Ranvier. Explicit expressions for the dyadic rim propagator and dominant near-zone field components were derived. The formulation reproduces the conventional quasistatic dipole limit while simultaneously identifying the underlying spectral sector as overwhelmingly evanescent.

The resulting framework refines the microscopic electrodynamic interpretation of ephaptic and extracellular interaction by explicitly characterizing the dominant biological electromagnetic sector as localized, longitudinally structured, and matter-attached.

A Appendix A: Interaction Energy and Thermal Scale Estimates

To assess possible biological relevance of the near-field interaction, consider two neighboring ion-channel dipoles:

$$\mathbf{p}_1, \quad \mathbf{p}_2. \quad (59)$$

The dipole-dipole interaction energy is:

$$U_{12} = -\mathbf{p}_2 \cdot \mathbf{E}_1. \quad (60)$$

Using the quasistatic near field:

$$\mathbf{E}_1 = \frac{1}{4\pi\epsilon_0} \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{p}_1) - \mathbf{p}_1}{r^3}, \quad (61)$$

one obtains:

$$U_{12} = -\frac{1}{4\pi\epsilon_0} \frac{3(\mathbf{p}_1 \cdot \hat{\mathbf{r}})(\mathbf{p}_2 \cdot \hat{\mathbf{r}}) - \mathbf{p}_1 \cdot \mathbf{p}_2}{r^3}. \quad (62)$$

For:

$$p_1 \sim p_2 \sim 10^{-24} \text{ C m}, \quad (63)$$

and:

$$r \sim 10^{-6} \text{ m}, \quad (64)$$

the interaction energy becomes:

$$U_{12} \sim 10^{-20} - 10^{-19} \text{ J.} \quad (65)$$

At physiological temperature:

$$k_B T \sim 4 \times 10^{-21} \text{ J.} \quad (66)$$

Thus:

$$\frac{U_{12}}{k_B T} \sim 1 - 10 \quad (67)$$

may occur in tightly packed geometries or during simultaneous multi-channel activation.

This estimate does not imply coherent quantum behavior. Rather, it suggests that microscopic near-field interactions may be energetically non-negligible relative to thermal fluctuations.

B Appendix B: Decoherence and Classicality

The present formulation does not require long-lived biological quantum coherence.

At:

$$T \sim 300 \text{ K,} \quad (68)$$

environmental decoherence is expected to occur rapidly through:

1. ionic collisions,
2. solvent interactions,
3. membrane fluctuations,
4. dielectric losses.

The characteristic ion-channel activation timescale is:

$$\tau_{\text{channel}} \sim 10^{-4} - 10^{-3} \text{ s.} \quad (69)$$

Electromagnetic decoherence times are expected to be substantially shorter.

Consequently, the quantized rim field should primarily be interpreted as:

1. a mathematically consistent quantization of the evanescent sector,
2. a propagator-theoretic refinement of near-field electrodynamics,
3. a microscopic interpretation of localized field-matter coupling.

The formalism therefore remains compatible with effectively classical biological behavior.

C Appendix C: Comparison with Conventional Ephaptic and Volume-Conductor Theory

Classical ephaptic models generally employ quasistatic extracellular potentials satisfying:

$$\nabla \cdot (\sigma \nabla \phi) = 0, \quad (70)$$

with:

$$\mathbf{E} = -\nabla \phi. \quad (71)$$

Finite-element and cable-theoretic approaches successfully reproduce:

1. extracellular voltage profiles,
2. local field coupling,
3. synchronization effects,
4. conduction perturbations.

The present rim-field formulation agrees with these models in the low-frequency limit. Indeed:

$$\omega \sim 10^4 \text{ s}^{-1} \quad (72)$$

implies:

$$k = \frac{\omega}{c} \sim 10^{-5} \text{ m}^{-1}. \quad (73)$$

At micron scales:

$$r \sim 10^{-6} \text{ m}, \quad (74)$$

we obtain:

$$kr \sim 10^{-11} \ll 1. \quad (75)$$

Hence the quasistatic approximation remains valid.

The novelty of the rim formulation is not the rejection of quasistatics but the identification of the dominant spectral sector as:

$$q \gg k, \quad (76)$$

corresponding to evanescent rather than propagating modes.

Thus the ordinary coordinate-space near field:

$$\mathbf{E}_{\text{near}} \propto \frac{1}{r^3} \quad (77)$$

is interpreted as the coordinate-space manifestation of an underlying evanescent spectral propagator.

D Appendix D: Relation to Nanoantenna Interpretations

Some previous studies have interpreted nodes of Ranvier or neural microstructures as nanoscale electromagnetic emitters or nanoantennas.

The present framework partially overlaps with such interpretations but differs fundamentally in one important respect.

At biological frequencies:

$$\lambda = \frac{2\pi c}{\omega} \sim 10^5 \text{ m}, \quad (78)$$

whereas biological dimensions satisfy:

$$r \sim 10^{-6} \text{ m}. \quad (79)$$

Thus:

$$r \ll \lambda. \quad (80)$$

Consequently:

$$q \gg k \quad (81)$$

throughout the biologically relevant spectral regime.

The node therefore behaves primarily as a localized near-field source rather than as an efficient radiative antenna.

E Appendix E: Possible Experimental Signatures

The rim-field framework suggests several experimentally testable consequences:

1. Enhanced ephaptic coupling in densely packed tracts,
2. Separation-dependent synchronization delays,
3. Geometry-dependent modulation of conduction velocity,
4. Strongly localized extracellular near-field gradients near nodes,
5. Increased coupling in partially demyelinated tissue.

Possible experimental approaches include:

1. nanoelectrode mapping near nodes of Ranvier,
2. controlled axon-separation experiments,
3. finite-element comparison with spectral evanescent decomposition,
4. high-resolution extracellular field imaging,
5. tract-geometry correlation studies.

F Appendix F: Structural Comparison with Keller’s Near-Field QED

The quantized rim-field formulation shares several structural similarities with Keller’s near-field quantum electrodynamics.

Present Rim-Field Formulation	Keller Near-Field QED
Evanescent spectral modes	Near-field photons
Localized rim-zone fields	Rim-zone localization
Longitudinal dominance	Scalar/longitudinal sector
Source-attached excitations	Photon embryos
Non-radiative localization	Matter-attached fields

The comparison is therefore:

1. structural,
2. propagator-theoretic,
3. spectral,
4. non-speculative.

The present work does not claim experimentally established photon-embryo dynamics in neural tissue. Rather, it identifies a mathematical correspondence between localized evanescent electrodynamic excitations and Keller’s near-field QED structures.